

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

83000095 - Advanced Design Of Naval And Offshore Structures

DEGREE PROGRAMME

08IN - Master Universitario En Ingenieria Naval Y Oceanica

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 2

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1. Description

1.1. Subject details

Name of the subject	83000095 - Advanced Design Of Naval And Offshore Structures
No of credits	6 ECTS
Type	Optional/elective
Academic year of the programme	Second year
Semester of tuition	Semester 4
Tuition period	February-June
Tuition languages	English
Degree programme	08IN - Master Universitario en Ingeniería Naval y Oceanica
Centre	08 - E.T.S. De Ingenieros Navales
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Jaime Moreu Gamazo		jaime.moreu@upm.es	Sin horario.
Mario De Vicente Peño		mario.devicente@upm.es	Sin horario.
Miguel Jesus Taboada Gosálvez		miguel.taboada.gosalvez@upm.es	Sin horario.
Arturo Silva Campillo (Subject coordinator)		a.silva@upm.es	- -

Miguel Angel Herreros Sierra		miguelangel.herreros@upm.es	Sin horario.
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* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Skills and learning outcomes *

3.1. Skills to be learned

CG1 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con su área de estudio. Students should be able to apply their acquired knowledge and problem-solving skills in new or unfamiliar environments within broader (or multidisciplinary) contexts related to their area of study.

CG2 - Que los estudiantes sean capaces de integrar conocimientos y enfrentarse a la complejidad de formular juicios a partir de una información que, siendo incompleta o limitada, incluya reflexiones sobre las responsabilidades sociales y éticas vinculadas a la aplicación de sus conocimientos y juicios. That students are able to integrate knowledge and face the complexity of making judgments based on incomplete or limited information, including reflections on the social and ethical responsibilities linked to the application of their knowledge and judgments.

CG3 - Que los estudiantes sepan comunicar sus conclusiones- y los conocimientos y razones últimas que las sustentan a públicos especializados y no especializados de un modo claro y sin ambigüedades. That students know how to communicate their conclusions - and the ultimate knowledge and reasons that support them - to specialized and non-specialized audiences in a clear and unambiguous manner.

CG4 - (S1) Que los estudiantes posean las habilidades de aprendizaje que les permitan continuar estudiando de un modo que habrá de ser en gran medida autodirigido o autónomo.

CTUPM01 - (S2) Creatividad. Los estudiantes deben resolver de forma nueva, original y aportando valor, situaciones o problemas en el ámbito de la ingeniería. Creativity. Students must solve situations or problems in the field of engineering in a new, original and valuable way.

CTUPM04 - (S5) Uso de la lengua inglesa. Los estudiantes establecen conversaciones con nativos sin tener problemas de comunicación adicionales tanto de forma oral como escrita. Use of the English language. Students establish conversations with native speakers without having additional communication problems both orally and in writing.

CTUPM05 - (S6) Uso de las tecnologías de la información y comunicación (TIC). Los estudiantes aplican conocimientos tecnológicos necesarios de manera que les permitan desenvolverse cómodamente y afrontar los retos que la sociedad les va a imponer en su quehacer profesional empleando la informática. Use of information and communication technologies (ICT). Students apply the necessary technological knowledge in a way that allows them to perform comfortably and meet the challenges that society will impose on them in their professional work using information technology.

CTUPM09 - Resolución de problemas. Los estudiantes son capaces de identificar o proponer un problema, y tienen el conocimiento sobre diferentes alternativas metodológicas y estratégicas para resolverlo. Problem solving. Students are able to identify or propose a problem, and have the knowledge about different methodological and strategic alternatives to solve it.

CTUPM13 - Trabajo en contextos internacionales. Los estudiantes son capaces de integrarse en un grupo o equipo, colaborando y cooperando con otros. Tienen la capacidad para trabajar con estudiantes de otras disciplinas y de aceptar la diversidad social y cultural. Work in international contexts. Students are able to integrate into a group or team, collaborating and cooperating with others. They have the ability to work with students from other disciplines and to accept social and cultural diversity.

3.2. Learning outcomes

RA4 - Competencias: Capacidad para el análisis dinámico determinista de estructuras en el dominio del tiempo y de la frecuencia

RA3 - Habilidades y Destrezas: Aplicación del Método de los Elementos Finitos para el proyecto de Estructuras Navales

RA5 - Habilidades y Destrezas: Aplicación del Método de los Elementos Finitos para el proyecto de Estructuras Navales

RA2 - Competencias: Capacidad para el análisis de estructuras navales y su optimización

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

4. Brief description of the subject and syllabus

4.1. Brief description of the subject

The course offers comprehensive training in the analysis, design, and cost estimation of marine and offshore structures, with a weekly commitment of 4 hours. It is organized into three main blocks: "Optimal Analysis of Marine Structures," which teaches advanced techniques for achieving optimal performance and safety; "Advanced Fatigue Design," focusing on understanding and mitigating fatigue in structural integrity and "Nonlinear Structural Analysis," which covers sophisticated tools and simulations for predicting nonlinear behavior.

4.2. Syllabus

1. Optimal analysis of marine structures
2. Advanced fatigue design
3. Nonlinear structural analysis

5. Schedule

5.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Optimal analysis of marine structures Duration: 04:00 Lecture			
2	Optimal analysis of marine structures Duration: 04:00 Lecture			
3	Optimal analysis of marine structures Duration: 04:00 Lecture			
4	Optimal analysis of marine structures Duration: 04:00 Lecture			
5	Optimal analysis of marine structures Duration: 02:00 Lecture			Midterm exam Written test Progressive assessment Presential Duration: 02:00
6	Advanced fatigue design Duration: 04:00 Lecture			
7	Advanced fatigue design Duration: 04:00 Lecture			
8	Advanced fatigue design Duration: 04:00 Lecture			
9	Advanced fatigue design Duration: 04:00 Lecture			
10	Advanced fatigue design Duration: 02:00 Lecture			Midterm exam Written test Progressive assessment Presential Duration: 02:00
11	Nonlinear structural analysis Duration: 04:00 Lecture			
12	Nonlinear structural analysis Duration: 04:00 Lecture Nonlinear structural analysis Duration: 04:00 Lecture			

13	Nonlinear structural analysis Duration: 04:00 Lecture			
14	Nonlinear structural analysis Duration: 04:00 Lecture			
15	Nonlinear structural analysis Duration: 04:00 Lecture			
16	Nonlinear structural analysis Duration: 02:00 Lecture			Midterm exam Written test Progressive assessment Presential Duration: 02:00
17				Final exam Written test Global examination Presential Duration: 04:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

6. Activities and assessment criteria

6.1. Assessment activities

6.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
5	Midterm exam	Written test	Face-to-face	02:00	%	/ 10	
10	Midterm exam	Written test	Face-to-face	02:00	%	/ 10	CTUPM01 CG2 CG3 CTUPM05 CTUPM09 CTUPM13 CG4 CTUPM04
16	Midterm exam	Written test	Face-to-face	02:00	25%	3 / 10	CG1 CTUPM01 CG2 CG3 CTUPM05 CTUPM09 CTUPM13 CG4 CTUPM04

6.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Final exam	Written test	Face-to-face	04:00	100%	5 / 10	CTUPM09 CTUPM13 CG1 CTUPM05 CTUPM04 CTUPM01 CG2 CG3 CG4

6.1.3. Referred (re-sit) examination

No se ha definido la evaluación extraordinaria.

6.2. Assessment criteria

The course is divided into four modules, which can be passed through progressive evaluation by calculating the arithmetic mean of each module with a grade of five or higher (out of 10). The minimum grade for each module is 3 (out of 10). Alternatively, students have the option to pass the course through a final exam that counts for 100% of the grade, with no option to carry over any previous module grades, and must achieve a score of five or higher (out of 10).

7. Teaching resources

7.1. Teaching resources for the subject

Name	Type	Notes
libro 1	Bibliography	Dinámica de estructuras autor: rodriguez rubio, angel editorial: etsi navales
libro 2	Bibliography	Dinámica de estructuras autor: espinosa de los monteros, ignacio editorial: etsi navales 1981
libro 3	Bibliography	Dynamics of structures: theory and applications autor: a. chopra editorial: prentice hall, 2000 isbn: 0130869732
libro 4	Bibliography	Dynamics of structures autor: raymond w. clough, joseph penzien editorial computers and structures, inc. 2nd edition, february 2004, isbn: 0-923907-50-5.
libro 5	Bibliography	Twelve lectures on structural dynamics. autor: andré preumont ed. universitate libre de bruxelles

libro 6	Bibliography	Dinámica estructural. teoría y cálculo. autor: mario paz. ed. reverté 1992 (reedición 2002)
libro 7	Bibliography	E. oñate, cálculo de estructuras por el método de los elementos finitos. 1- análisis estático lineal, 2- análisis no lineal, cimne, 1992.
libro 8	Bibliography	Fundamentos de Dinámica Estructural de Enrique de la Fuente, ed Garceta
material Moodle	Web resource	Material en la plataforma Moodle de la asignatura